

Question Bank, PT-2

AS-I

Q1

Question

Calculate the Mean, Standard Deviation, and Coefficient of Variation for the following distribution:

Class Interval	Frequency (f)
0-10	5
10-20	9
20-30	12
30-40	8
40-50	6

Q2

A discrete random variable X has PMF:

$$P(X = x) = kx, \quad x = 1, 2, 3, 4$$

Find

1. k
2. $P(X \leq 2)$
3. CDF of X

Q3

A random variable X has the following probability distribution

X	:	0	1	2	3	4	5	6	7
$P_X(x)$	:	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

Then find the following:

- (i) k , (ii) $P(0 < X < 5)$, (iii) $P(X < 6)$, (iv) $P(X \geq 6)$ (v) CDF

Q4

Find the SD and variance of the following data

Size of item:	10	11	12	13	14	15	16
Frequency:	2	7	11	15	10	4	1

Q5

Find the probability that at most 5 defective fuses will be found in a box of 200 fuses if 2% of such fuses are defective.

Q6 (a)

A random variable X is uniformly distributed over integers from 3 to 10.

Find:

(a) Mean

(b) Variance

(b)

A random variable X takes values $\{1, 2, 3, 4, 5, 6\}$ with equal probability.

Find:

(a) PMF

(b) Mean

(c) Variance

Q7

Calculate the first four moments of the following distribution about the mean and hence find the γ_1 and γ_2 .

$x :$	0	1	2	3	4	5	6	7	8
$f :$	1	8	28	56	70	56	28	8	1

Q8 Given

$$p(x, y) = k(x + y)$$

for

$$x = 0, 1$$

$$y = 0, 1, 2$$

Find:

1. k
2. Marginal distributions

Q9

Joint PMF:

$X \backslash Y$	0	1
0	0.1	0.2
1	0.3	0.4

Find:

1. $P(X = 1 \mid Y = 1)$
 2. $P(Y = 0 \mid X = 0)$
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Q10

Let the joint PMF be:

$$p(x, y) = k(x + 2y), \quad x = 0, 1, 2; \quad y = 0, 1$$

Tasks:

1. Find the value of k
2. Find marginal PMFs $P_X(x)$ and $P_Y(y)$
3. Check whether X and Y are independent
4. Find $P(X + Y \geq 2)$
5. Find conditional probability $P(X = 2 \mid Y = 1)$